THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS UNIT TEST SET A

Chapters 12–14: **Surface Areas and Volumes, Statistics, Probability** | RBSE Board Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat figures; use graph paper where a graph is asked.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. Two cubes each of edge 5 cm are joined end to end. The surface area of the resulting cuboid is: 1

(a) 150 cm² **(b)** 200 cm² **(c)** 250 cm² **(d)** 300 cm²

2. When a hemisphere is placed on a cylinder of the same radius, the surfaces not counted in the total surface area are: 1

(a) the bottom of the cylinder
(b) the curved surfaces
(c) the base of the hemisphere and the top of the cylinder
(d) none

3. A toy is a cone (radius 3 cm, slant height 5 cm) on a hemisphere of radius 3 cm. Student A says its surface area is $\pi rl + 2\pi r^2 = 33\pi$; Student B says $\pi rl + 3\pi r^2 = 42\pi$. Then: 1

(a) Student B is correct **(b)** Student A is correct
(c) Both are correct **(d)** Both are wrong

4. The curved surface area of a cone of radius 7 cm and slant height 10 cm is: 1

(a) 110 cm² **(b)** 440 cm² **(c)** 220 cm² **(d)** 154 cm²

5. If the mean is 20 and the median is 22, then by the empirical relation the mode is: 1

(a) 18 **(b)** 24 **(c)** 26 **(d)** 22

6. The cumulative frequency of a class is: 1

(a) the class mark × frequency
(b) its own frequency
(c) the sum of frequencies up to and including that class
(d) half the total

7. A student states: "The median of grouped data is the mid-point of the middle class." The correct response is: 1

(a) Correct **(b)** Incorrect — use $l + \frac{\frac{n}{2} - cf}{f} \times h$
(c) Correct for odd n **(d)** Cannot be determined

8. In the step-deviation method, $u = \frac{x - a}{h}$. If $a = 25$, $h = 10$ and $x = 45$, then u is: 1

(a) 20 **(b)** −2 **(c)** 2 **(d)** 0.2

9. Which of the following can be the probability of an event? 1

(a) −0.5 **(b)** 1.01 **(c)** 0.99 **(d)** $\frac{3}{2}$

10. If $P(E) = 0.35$, then $P(\text{not } E)$ is: 1

(a) 1.35 **(b)** 0.53 **(c)** 0.35 **(d)** 0.65

11. A coin is tossed twice. The probability of getting exactly one head is: 1

(a) $\frac{3}{4}$ **(b)** $\frac{1}{2}$ **(c)** $\frac{1}{4}$ **(d)** 1

12. A student states: "If $P(A) = 0.7$, then $P(\text{not } A) = 0.7$." The correct response is: 1

(a) Correct **(b)** Incorrect; $P(\text{not } A) = 0.3$

(c) Correct only if A is sure **(d)** Cannot be determined

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

13. A cylinder of radius 7 cm and height 10 cm has a hemisphere of the same radius on top. Find the total volume of the solid. 2

14. Two cubes, each of volume 27 cm³, are joined end to end. A student says the volume of the cuboid is 27 cm³. Justify or refute, and find its surface area. 2

15. Find the mode of: 2

Class	0–20	20–40	40–60	60–80
Frequency	15	6	18	10

16. A student says the mean of the data below is 30 (the average of the class marks). Justify or refute. 2

Class	10–20	20–30	30–40	40–50
Frequency	2	4	6	8

17. Harpreet tosses two different coins simultaneously. Find the probability that she gets 2

(i) at least one tail

(ii) two heads.

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 40–60 words each.

18. A cubical block of side 7 cm is surmounted by a hemisphere. What is the greatest diameter the hemisphere can have? Find the surface area of the solid. 3

19. From a solid cylinder of height 2.4 cm and diameter 1.4 cm, a conical cavity of the same height and diameter is hollowed out. Find the total surface area of the remaining solid to the nearest cm². 3

20. Find the mode of the ages of patients: 3

Age (yrs)	5–15	15–25	25–35	35–45	45–55	55–65
Patients	6	11	21	23	14	5

21. A jar contains 24 marbles, some green and others blue. The probability of drawing a green marble is $\frac{2}{3}$. 3

(i) Find the number of blue marbles.

(ii) If 4 more blue marbles are added, find the new probability of a green marble.

22. A bowl is a hollow hemisphere of internal radius 10.5 cm with a hollow cylinder of the same radius and height 14 cm on top. Find its inner surface area. 3

OR

A cone of radius 3 cm and height 4 cm is hollowed out from a solid cylinder of radius 3 cm and height 7 cm. Find the volume of the remaining solid. 3

23. Find the median of: 3

Marks	0–10	10–20	20–30	30–40	40–50
Students	6	10	14	12	8

OR

Find the mean daily expenditure on food by the step-deviation method: 3

Expense (₹)	100–150	150–200	200–250	250–300	300–350
Households	4	5	12	2	2

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

24. **Case study:** A tent is a cylinder of radius 14 m and height 3 m surmounted by a cone of height 10.5 m. **5**

(i) Find the slant height.

(ii) Find the canvas needed.

(iii) Find its cost at ₹80 per m².

(iv) Find the volume of air inside.

OR

Case study: An ice-cream shop serves a cup that is a cylinder of radius 3 cm and height 5 cm, topped with a hemisphere of ice-cream of the same radius. (Use $\pi = 3.14$) **5**

(i) Find the volume of the cylindrical part.

(ii) Find the volume of the hemispherical top.

(iii) Find the total ice-cream in one cup.

(iv) How many litres are needed for 50 cups?

25. **Case study:** At a school fair, a bag has tokens numbered 1 to 30. A player wins if the token drawn is a multiple of 5 or a perfect square. **5**

(i) Find P(multiple of 5).

(ii) Find P(perfect square).

(iii) Find P(winning).

(iv) A student says P(winning) $= \frac{11}{30}$. Find the mistake.

OR

A bag contains 18 balls out of which x are red. **5**

(i) If one ball is drawn at random, find the probability that it is not red.

(ii) If 2 more red balls are put in the bag, the probability of drawing a red ball becomes $\frac{9}{8}$ times the earlier probability. Find x.

END OF QUESTION PAPER — Total: 50 Marks

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